MATHEMATICS

LOGARITHMS

Combined Test Paper with Answers

Name: _____ Class: _____ Date: _____

Questions: 153 **Total marks: 690**

READ THESE INSTRUCTIONS FIRST

- Answer all the questions.
- Show all necessary working clearly.
- Calculators may be used unless a question says non-calculator (Question 98).
- Give non-exact answers correct to 3 significant figures, or to the accuracy stated in the question.
- Model answers are printed in the shaded left-hand column beside each question.

INFORMATION

- The number of marks is given in brackets [ ] at the end of each question or part-question.
- $\log_b x$ means the logarithm of x to the base b. a/b means a divided by b. ∪ means union. ∞ means infinity. Brackets ( ) exclude an end value; square brackets [ ] include it.
- Sections A to F are standard practice questions. Sections G to L are exam questions in order of difficulty.
- Marks for Sections G to L have been assigned for this paper.

SECTION A: EVALUATING LOGARITHMS

Evaluate each of the following.

Answer	No.	Question
4	**1**	$\log_3 81$ **[2]**
4/3	**2**	$\log_{27} 81$ **[2]**
5/3	**3**	$\log_8 32$ **[2]**
5/6	**4**	$\log_{27}(9\sqrt{3})$ **[2]**
2	**5**	$\log_{1/3}(1/9)$ **[2]**
−3	**6**	$\log_{1/11} 1331$ **[2]**
6	**7**	$\log_{2\sqrt{3}} 1728$ **[2]**
−1/2	**8**	$\log_2(\cos 45°)$ **[2]**
−1	**9**	$\log_{2-\sqrt{3}}(2+\sqrt{3})$ **[2]**
−1	**10**	$\log_{0.125} 8$ **[2]**
−1	**11**	$\log_{3/4}(1.333\ldots)$ (recurring decimal; 1.333… = 4/3) **[2]**
−1	**12**	$\log_{2.25}(0.444\ldots)$ (recurring decimal; 0.444… = 4/9) **[2]**
0	**13**	$\log_{10}(0.999\ldots)$ (recurring decimal; 0.999… = 1) **[2]**
12	**14**	$\log_{\sqrt[3]{7}} 2401$ (the base is the cube root of 7) **[3]**
0	**15**	$\log_2 \log_2 \log_3 \log_3(27^3)$ **[3]**
1	**16**	$\log_5\sqrt{(5\sqrt{(5\sqrt{(5\ldots)})})}$ (the square roots go on for ever) **[3]**

SECTION B: PROPERTIES OF LOGARITHMS

Answer	No.	Question
1	**17**	Simplify $\log_a b \times \log_b a$. **[2]**
1	**18**	Simplify $\log_a b \times \log_b c \times \log_c d \times \log_d e \times \log_e a$. **[2]**
n = 1023	**19**	$\log_2 3 \times \log_3 4 \times \log_4 5 \times \ldots \times \log_n(n+1) = 10$. Find n. **[4]**
3/2	**20**	$A = \log_{11}(11^{\log_{11} 1331})$ $B = \log_{385} 5 + \log_{385} 7 + \log_{385} 11$ $C = \log_4(\log_2(\log_5 625))$ $D = 10^{\log_{100} 16}$ Find the value of $(A \times B)/(C \times D)$. **[6]**

Answer	No.	Question
N = 2 (a prime number, co-prime with 7)	**21**	If $N = \log_3 12 / \log_{36} 3 - \log_3 4 / \log_{108} 3$, then N is: (i) a composite number (ii) a prime number (iii) co-prime with 7 (iv) a rational number that is not an integer **[4]**

SECTION C: LOGARITHMIC EQUATIONS

Answer	No.	Question
x = 1	**22**	Find x: $x^2 + 7^{\log_7 x} - 2 = 0$ **[3]**
4	**23**	Find the sum of all solutions of $\log_{10} x + \log_{10}(x+2) - \log_{10}(5x+4) = 0$ **[4]**
x = 9 or x = 1/9	**24**	Find x: $3^{(\log_3 x)^{\{2\}}} + x^{\log_3 x} = 162$ **[5]**
x = 41	**25**	Find x: $\log_{10}(2^x + x - 41) = x(1 - \log_{10} 5)$ **[4]**
1	**26**	Find the product of all solutions of $x^{1+\log_{10} x} = 100000x$ **[5]**
x = 99 or x = −9/10	**27**	Find x: $(x+1)^{\log_{10}(x+1)} = 100(x+1)$ **[4]**
x = 1/4 and x = 1/2	**28**	Find x: $\log_5(5^{1/x} + 125) = \log_5 6 + 1 + 1/(2x)$ **[5]**
No solution	**29**	Solve for x: $\log_2(9 - 2^x) / (x - 3) = 1$ **[3]**
x = 3 and x = 5/4	**30**	Solve for x: $\log_{x-1} 4 = 1 + \log_2(x - 1)$ **[5]**
8	**31**	Find the value of: $[(\log_2 9)^2]^{1/\log_2(\log_2 9)} \times (\sqrt{7})^{1/\log_4 7}$ **[5]**
x = ln 4/(ln 4 − ln 3) ≈ 4.82; options A, B and C	**32**	If $3^x = 4^{x-1}$, then x = ? (A) $2\log_3 2 / (2\log_3 2 - 1)$ (B) $2 / (2 - \log_2 3)$ (C) $1 / (1 - \log_4 3)$ (D) $2\log_2 3 / (2\log_2 3 - 1)$ **[4]**

SECTION D: PRACTICE SET: SOLVE

Solve each of the following equations.

Answer	No.	Question
x = 1 + √3	**33**	(i) $\log_{x-1} 3 = 2$ **[4]**
x = 3	**34**	(ii) $\log_4(2\log_3(1 + \log_2(1 + 3\log_3 x))) = 1/2$ **[4]**
x = 4	**35**	(iii) $\log_3(1 + \log_3(2^x - 7)) = 1$ **[4]**
x = 2	**36**	(iv) $\log_3(3^x - 8) = 2 - x$ **[4]**
x = 0 (x = 3 is rejected)	**37**	(v) $\log_2(9 - 2^x) / (3 - x) = 1$ **[4]**
x = −5	**38**	(vi) $\log_{5-x}(x^2 - 2x + 65) = 2$ **[4]**

Answer	No.	Question
$x = 3/2$ or $x = 10$	**39**	(vii) $\log_{10}5 + \log_{10}(x + 10) - 1 = \log_{10}(21x - 20) - \log_{10}(2x - 1)$ **[4]**
$x = 10$ or $x = 1/10$	**40**	(viii) $x^{1+\log_{10}x} = 10x$ **[4]**
$x = \sqrt{5}$ or $x = 5$	**41**	(ix) $2(\log_x\sqrt{5})^2 - 3\log_x\sqrt{5} + 1 = 0$ **[4]**
$x = 8$ or $x = 1/\sqrt{3} - 1 \approx -0.423$	**42**	(x) $3 + 2\log_{x+1}3 = 2\log_3(x + 1)$ **[4]**

SECTION E: LOGARITHMIC INEQUALITIES (CONSTANT BASE)

Answer	No.	Question
$x > 1/8$	**43**	Solve $\log_{1/2}x < 3$. **[3]**
$x > 10$	**44**	Solve $\log_3(x - 1) > 2$. **[3]**
$(1, \infty)$	**45**	Find the domain of $f(x) = \log_{10}(x - 1)$. **[2]**
$(-\infty, 3)$	**46**	Find the domain of $f(x) = \log_2(3 - x)$. **[2]**
$(1/5, 2/5)$	**47**	Solve $\log_{1/3}(5x - 1) > 0$. **[4]**
$(1, 2) \cup (3, 4)$	**48**	Solve $\log_{0.5}(x^2 - 5x + 6) > -1$. **[4]**
$(-1, 0) \cup (1, 2)$	**49**	Solve $\log_{1/4}(2 - x) > \log_{1/4}(2/(x+1))$. **[5]**
$(-4, -3) \cup (8, \infty)$	**50**	Solve $\log_{0.5}[\log_6((x^2 + x)/(x + 4))] < 0$. **[5]**
$[-4, -3) \cup (0, 1]$	**51**	Find the solution set of $2 - \log_2(x^2 + 3x) \geq 0$. **[4]**
$(-\infty, -2) \cup (5/8, \infty)$	**52**	If $\log_{1/3}[(3x - 1)/(x + 2)]$ is less than 1, then x lies in: **[4]**
$(1/16, 1/8) \cup (8, 16)$	**53**	Solve $(\log_2x)^4 - [\log_{1/2}(x^{5/4})]^2 - 20\log_2x + 148 < 0$. **[6]**
$(2, \infty)$	**54**	If $\log_{0.3}(x - 1) < \log_{0.09}(x - 1)$, then x lies in the interval: **[4]**
$[-1/2, -1/4) \cup (3/4, 1]$	**55**	Solve $\log_{5/8}[2x^2 - x - 3/8] \geq 1$. **[5]**

SECTION F: LOGARITHMIC INEQUALITIES (VARIABLE BASE)

Answer	No.	Question
$(3/8, 1/2) \cup (1, 3/2)$	**56**	Solve $\log_x(2x - 3/4) > 2$. **[5]**
$(2, \infty)$	**57**	Solve $\log_x(x^3 - x^2 - 2x) < 3$. **[5]**
$(0, 1/2) \cup (1, 2) \cup (3, 6)$	**58**	Solve $\log_{2x}(x^2 - 5x + 6) < 1$. **[5]**
$(-2, -1) \cup (-1, 0) \cup (0, 1) \cup (2, \infty)$	**59**	Solve $\log_{x^2}(2 + x) < 1$. **[5]**
$(-6, -5) \cup (-3, -2)$	**60**	Solve $\log_{(x+6)/3}[\log_2((x - 1)/(x + 2))] > 0$. **[6]**

SECTION G: EXAM QUESTIONS (**, **+)

Answer	No.	Question
Shown: LHS = $\log_a(1/4) = -\log_a 4$	**61**	Show clearly that $\log_a 36 + \frac{1}{2}\log_a 256 - 2\log_a 48 = -\log_a 4$. **[3]**
3	**62**	Simplify $\log_2 5 + \log_2 1.6$, giving the final answer as an integer. **[2]**
Shown: $\log_2(2^{3p+2q}) = 3p + 2q$	**63**	Given $x = 2^p$ and $y = 4^q$, show clearly that $\log_2(x^3y) = 3p + 2q$. **[3]**
(a) −3; (b) 6	**64**	Simplify each of the following expressions, giving the final answer as an integer. Full workings, justifying every step, must support each answer. (a) $\log_2 3 - \log_2 24$ **[2]** (b) $\log_a a^2 - 4\log_a(1/a)$, $a > 0, a \neq 1$ **[3]**
(a) 2y; (b) 3 + 2y	**65**	Given that $\log_2 x = y$, write each of the following in terms of y. (a) $\log_2 x^2$ **[1]** (b) $\log_2(8x^2)$ **[2]**
$x = \frac{1}{2}\log_{10}(y/4)$	**66**	Given that $y = 4 \times 10^{2x}$, express x in terms of y, giving an exact simplified answer in terms of logarithms base 10. **[3]**
$x = (\log y - \log a)/\log b$	**67**	An exponential curve has equation $y = ab^x$, $x \in \mathbb{R}$, where a and b are non-zero constants. Make x the subject of the equation, giving the final answer in terms of logarithms base 10. **[3]**
x = 5	**68**	Solve the logarithmic equation $2\log_{10} x + \log_{10} 3 = \log_{10} 75$. **[3]**
x = 5	**69**	Solve the logarithmic equation $\log_a x + \log_a(x - 3) = \log_a 10$. **[3]**

SECTION H: EXAM QUESTIONS (***)

Answer	No.	Question
(a) decreasing curve through (0, 1), asymptote y = 0; (b) x ≈ 0.369	**70**	An exponential curve C has equation $y = 1/3^x$, $x \in \mathbb{R}$. (a) Sketch the graph of C. **[2]** (b) Solve the equation y = 2/3, giving the answer correct to 3 significant figures. **[3]**
(a) p + 2q; (b) ½p − q	**71**	Given that $\log_a 4 = p$ and $\log_a 5 = q$, express each of the following in terms of p and q. The final answers may not contain any logarithms. (a) $\log_a 100$ **[2]** (b) $\log_a 0.4$ **[2]**

Answer	No.	Question
$t = 1/3$	**72**	Solve the logarithmic equation $\log_5(4t + 7) - \log_5 t = 2$. **[4]**
(a) $2p + q$; (b) $p - q - 1$	**73**	Given that $\log_2 3 = p$ and $\log_2 5 = q$, express each of the following in terms of p and q. The final answers may not contain any logarithms. (a) $\log_2 45$ **[2]** (b) $\log_2 0.3$ **[3]**
(a) $x \approx 1.18$; (b) $y = 8$ or $y = 1/8$	**74**	Solve each of the following equations, giving the final answers correct to 3 significant figures where appropriate. (a) $7^x = 10$ **[2]** (b) $\log_2 y = 9\log_y 2$ **[3]**
$x = 10$	**75**	Solve the following logarithmic equation for x. $\log_a(x^2 - 10) - \log_a x - 2\log_a 3 = 0$ **[4]**
$x = 6$ or $x = 12$	**76**	Solve the following logarithmic equation for x. $2\log_a x = \log_a 18 + \log_a(x - 4)$ **[4]**
$z = 1/2$	**77**	Solve the following logarithmic equation. $\log_2(2z + 1) = 2 + \log_2 z$ **[4]**
$y = 8$ or $y = 12$	**78**	Solve the following logarithmic equation for y. $2\log_a y - \log_a(5y - 24) = \log_a 4$ **[4]**
(a) $x = k^2/4$; (b) $y = (3x^4 - 1)/5$	**79**	It is given that $\log_a x = 2(\log_a k - \log_a 2)$, where $k > 0$, $a > 0$, $a \neq 1$. (a) Find x in terms of k, giving the answer in a form not involving logarithms. **[2]** (b) Suppose instead that $\log_x(5y + 1) = 4 + \log_x 3$, where $x > 0$, $x \neq 1$. Solve this equation for y in terms of x, in a form not involving logarithms. **[3]**
$x = 5$	**80**	Solve the logarithmic equation $\log_5(125x) = 4$. **[3]**
$x = 1/5$ or $x = 3$	**81**	Solve the logarithmic equation $1 + 2\log_5 x = \log_5(16x - 3)$. **[4]**
$t \approx 14.2$ years	**82**	Every £1 invested in a saving scheme gains interest at the rate of 5% per annum, so that the total value of this £1 investment after t years is £y. This is modelled by the equation $y = 1.05^t$, $t \geq 0$. Find after how many years the investment will double. **[3]**
(a) $x = 4/3$; (b) $y = 3/2$	**83**	Solve each of the following logarithmic equations. (a) $\log_x 16 = \log_x 9 + 2$ **[3]** (b) $\log_y 27 = 3 + \log_y 8$ **[3]**

Answer	No.	Question
(a) $x \approx 5.56$; (b) $y = 7$	**84**	Solve each of the following equations, giving the final answers correct to 3 significant figures where appropriate. (a) $2 \times 3^x = 900$ **[3]** (b) $\log_2(7y - 1) = 3 + \log_2(y - 1)$ **[3]**

SECTION I: EXAM QUESTIONS (***+)

Answer	No.	Question
$\log_n(36n)$	**85**	Simplify fully $1 + 2\log_n 3 + \log_n 4$, giving the final answer as a single logarithm. **[3]**
$w = 1/5$	**86**	Solve the following logarithmic equation. $\log_2(w^2 + 4w + 3) = 4 + \log_2(w^2 + w)$ **[4]**
$x = \log_2 6$	**87**	Solve the following exponential equation, giving the answer as a single logarithm of base 2. $(1/2)^x = 1/6$ **[3]**
$x = 4$, $y = 1/2$	**88**	Solve the following simultaneous logarithmic equations. $\log_2(xy^2) = 0$ $\log_2(x^2y) = 3$ **[5]**
$t = 21$	**89**	Solve the following logarithmic equation. $2\log_3 t = 1 + \log_3 7t$ **[4]**
$t = 2/3$	**90**	Solve the following logarithmic equation. $\log_3 8 - 3\log_3 t = 3$ **[4]**
$w = 4/5$	**91**	Solve the following logarithmic equation. $\log_5(4 - w) - 2\log_5 w = 1$ **[4]**
$y = 4$	**92**	Solve the following logarithmic equation. $\log_2 y + \log_2(3y + 4) = 2\log_2(3y - 4)$ **[5]**
$x = 2$ or $x = 4$	**93**	Solve the following logarithmic equation. $\log_2(6 - x) = 3 - \log_2 x$ **[4]**
$x = 16$	**94**	Solve the following logarithmic equation. $\log_4 x = \log_3 9$ **[3]**
$A = 9500$, $k \approx 0.10$	**95**	The population P of a certain town in time t years is modelled by the equation $P = A \times 10^{kt}$, $t \geq 0$, where A and k are non-zero constants. When $t = 3$, $P = 19000$ and when $t = 6$, $P = 38000$. Find the value of A and the value of k, correct to 2 significant figures. **[5]**
$x = 3$ or $x = 6$	**96**	Solve the following logarithmic equation. $2\log_3 x - \log_3(x - 2) = 2$ **[4]**

Answer	No.	Question
$y = \sqrt[3]{(1/100)}$	**97**	Given that $x \neq 0$ and $y \neq 0$, and that logarithms are to base 10, $2\log(x/y) - 1 = \log(10x^2y)$, find the exact value of y. **[5]**
2	**98**	(Non calculator) The points P and Q lie on the curve with equation $y = 6\log_2 x - \log_2 7$, $x > 0$. The x coordinates of P and Q are 3 and 6 respectively. Find the gradient of the straight line segment PQ. **[4]**
(a) vertical stretch, scale factor 3; (b) increasing curve through (0, 3), asymptote y = 0; (c) $x \approx -0.792$	**99**	(a) Describe the geometric transformation which maps the graph of $y = 2^x$ onto the graph of $y = 3 \times 2^x$. **[1]** (b) Sketch the graph of $y = 3 \times 2^x$. **[2]** The curve with equation $y = 2^{-x}$ intersects the curve with equation $y = 3 \times 2^x$ at the point P. (c) Determine, correct to 3 decimal places, the x coordinate of P. **[3]**
(a) $p + 3q$; (b) $4q - \frac{1}{2}p$; (c) $p - q + 1$	**100**	It is given that $\log_6 25 = p$ and $\log_6 2 = q$. Express in terms of p and q each of the following expressions. (a) $\log_6 200$ **[2]** (b) $\log_6 3.2$ **[2]** (c) $\log_6 75$ **[3]**

SECTION J: EXAM QUESTIONS (****)

Answer	No.	Question
(a) y-intercept (0, 1/3), asymptote y = 0; (b) $x \approx 3.10$; (c) $x \approx 2.71$	**101**	Consider the curve $y = 3^{x-1}$, $x \in \mathbb{R}$. (a) Sketch the graph of $y = 3^{x-1}$, showing the coordinates of all intercepts with the coordinate axes. **[3]** (b) Find to 3 significant figures the x coordinate of the point where the curve meets the straight line y = 10. **[2]** (c) Determine to 3 significant figures the x coordinate of the point where the curve meets the curve $y = 2^x$. **[3]**
x = 4	**102**	Solve the following logarithmic equation, for $x > 0$. $16\log_2 x + 4\log_4 x + 2\log_{16} x = 37$ **[5]**
(a) A = 100; (b) 2019	**103**	In 1970 the average weekly pay of footballers in a certain club was £100. The average weekly pay, £P, is modelled by the equation $P = A \times b^t$, where t is the number of years since 1970, and A and b are positive constants. In 1991 the average weekly pay of footballers in the same club had risen to £740. (a) Find the value of A and show that b = 1.10, correct to three significant figures. **[4]** (b) Determine the year when the average weekly pay of footballers in this club will first exceed £10000. **[4]**

Answer	No.	Question
(a) $x \approx -0.0339$; (b) $y = 11/28$; (c) $t \approx 0.528$	**104**	Solve each of the following equations, giving the final answers correct to three significant figures, where appropriate. (a) $6^{3x+2} = 30$ **[3]** (b) $\log_4(12y + 5) - \log_4(1 - y) = 2$ **[4]** (c) $8^{2t} - 8^t - 6 = 0$ **[4]**
Shown: LHS $= \log_5 8 = 3\log_5 2$	**105**	Show clearly that $\log_5 6 + 2\log_5 2 - \log_{25} 9 = 3\log_5 2$. **[4]**
$x = 4$	**106**	Solve the following logarithmic equation. $\log_{10}(x + 4) + \log_{10}(x + 16) = 1 + 2\log_{10} x$ **[5]**
7/6	**107**	Simplify $\log_4 8 - \log_{27} 3$, giving the final answer as a simplified fraction. **[3]**
$3a/4 + 1$	**108**	Given that $\log_b 16 = a$, express $\log_b(8b)$ in terms of a. **[3]**
(i) $p + 3q$; (ii) $4q - \frac{1}{2}p$; (iii) $p - q + 1$	**109**	It is given that $\log_6 25 = p$ and $\log_6 2 = q$. Simplify each of the following logarithms, giving the final answers in terms of p, q and positive integers, where appropriate. (i) $\log_6 200$ **[2]** (ii) $\log_6 3.2$ **[2]** (iii) $\log_6 75$ **[3]**
$x \approx 1.58$	**110**	Solve the following exponential equation, giving the answer correct to 3 significant figures. $2^{2x} - 2^x - 6 = 0$ **[4]**
Shown: $x = 1/(3 - \log_2 3)$	**111**	Two curves C_1 and C_2 are defined for all values of x and have respective equations $y = 8^x$ and $y = 2 \times 3^x$. Show that the x coordinate of the point of intersection of the two curves is given by $1/(3 - \log_2 3)$. **[5]**
$x = \sqrt{2}$	**112**	Solve the following logarithmic equation. $\log_2 x = \log_4 2$ **[3]**
(a) (0, 2), ($\log_2 3$, 0), asymptote $y = -1$; (b) $-1 < f(x) \leq 2$; (c) $3/x - 1$	**113**	The functions f and g are defined as $f(x) = 3 \times 2^{-x} - 1$, $x \in \mathbb{R}$, $x \geq 0$, and $g(x) = \log_2 x$, $x \in \mathbb{R}$, $x \geq 1$. (a) Sketch the graph of f. Mark clearly the exact coordinates of any points where the curve meets the coordinate axes, in exact form in terms of logarithms base 2, and mark and label the equation of the asymptote. **[4]** (b) State the range of f. **[2]** (c) Find f(g(x)) in its simplest form. **[2]**
$x = 3\sqrt{3}$	**114**	Solve the following logarithmic equation. $\log_3 x = \log_9 27$ **[3]**

Answer	No.	Question
$a \approx 2.339$, $n \approx 2.096$	**115**	The points (2, 10) and (6, 100) lie on the curve with equation $y = ax^n$, where a and n are non-zero constants. Find, to three decimal places, the value of a and the value of n. **[5]**
$y \approx 2.32$	**116**	Solve the following exponential equation, giving the answer correct to 3 significant figures. $4^y - 3(2^y) - 10 = 0$ **[4]**
$x = 9$	**117**	Solve the following logarithmic equation. $2\log_3 x - \log_9 x = 3$ **[4]**
Shown: only solution is $x = 4$, $y = 8$	**118**	Show that $x = 4$ and $y = 8$ is the only solution pair of the following simultaneous logarithmic equations. $\log_2(3x + 4) = 1 + \log_2 y$ $2\log_2 y = 3\log_2 x$ **[6]**
(a) shown; (b) $a \approx 1.217$; (c) $t \approx 14.13$; (d) $P \to 500$ as $t \to \infty$	**119**	A population P of an endangered species of animals was introduced to a park. The population obeys the equation $P = 1000 / (2 + ka^{-t})$, $t \geq 0$, where k and a are positive constants, and t is the time in years since the species was introduced to the park. Initially 100 individual animals were introduced to the park, and this population doubled after 5 years. (a) Show that $k = 8$. **[2]** (b) Find the value of a, correct to 4 significant figures. **[3]** (c) Determine the value of t when $P = 400$. **[3]** (d) Explain why this population cannot exceed 500. **[2]**
$x = 162$	**120**	Solve the following logarithmic equation. $\log_3 x - \log_9(2x) = 2$ **[5]**
$x = 8$	**121**	Solve the following logarithmic equation. $\log_4 x - \log_{16}(x - 4) = 1$ **[5]**
$(x, y) = (2, 8)$ or $(8, 2)$	**122**	Solve the following simultaneous equations. $\log_2 x + 2\log_4 y = 4$ $x + y = 10$ **[5]**
$x = a^5$ or $x = a^{-4}$	**123**	Given that a is a positive constant greater than 1, solve the following logarithmic equation. $(\log_a x)^2 = \log_a x + 20$ **[5]**
Shown: $x = 1/(\log_2 7 - \log_2 5)$	**124**	Two curves C_1 and C_2 are defined for all values of x and have respective equations $y = 7^x$ and $y = 2 \times 5^x$. Show that the x coordinate of the point of intersection is given by $1/(\log_2 7 - \log_2 5)$. **[5]**
$t = 1$ or $t \approx 1.63$	**125**	Solve the following exponential equation, giving the answer correct to 3 significant figures where appropriate. $3^{2t-1} + 6 = 3^{t+1}$ **[5]**

Answer	No.	Question
$x = 4\sqrt{2}$, $y = \sqrt{2}$	**126**	Solve the following simultaneous logarithmic equations, giving the answers as exact simplified surds. $\log_y x = 5$ $\log_2 x = 2 + \log_2 y$ **[6]**
$x = 4$ or $x = 16$	**127**	Solve the following logarithmic equation, for $x > 0$, $x \neq 1$. $\log_4 x + \log_x 16 = 3$ **[6]**
$x = 2$, $y = 8$	**128**	Solve the following simultaneous logarithmic equations. $3\log_8(xy) = 4\log_2 x$ $\log_2 y = 1 + 2\log_2 x$ **[6]**
(a) translation 4 units right; (b) AB = 4; (c) $k = 16/3$	**129**	The figure (not reproduced here) shows the graphs of the curves with equations $y = \log_2 x$ and $y = \log_2(x - 4)$. The points A and B are the respective x intercepts of the two graphs. (a) Describe the geometric transformation which maps the graph of $y = \log_2 x$ onto the graph of $y = \log_2(x - 4)$. **[1]** (b) State the distance AB. **[1]** The straight line with equation $x = k$, where k is a positive constant, meets the graph of $y = \log_2 x$ at the point P and the graph of $y = \log_2(x - 4)$ at the point Q. (c) Given that the distance PQ is 2 units, determine the value of k. **[4]**
(a) 5 g; (b) T = 30; (c) N = 34	**130**	The radioactive decay of a phosphorus isotope is modelled by the equation $m = m_0 \times 2^{-0.2t}$, $t \geq 0$, where m is the mass of phosphorus left, in grams, and t is the time in days since the decay started. The initial mass of phosphorus is m_0. (a) Find the mass of phosphorus left when an initial mass of 20 grams is left to decay for 10 days, according to this model. **[2]** (b) An initial mass m_0 grams decays to $m_0/64$ grams in T days. Find the value of T. **[3]** (c) After N days have elapsed, less than 1% of this type of phosphorus remains from its initial mass m_0. Find the smallest integer value of N. **[3]**
$x = 1/4$ or $x = 16$	**131**	Solve the following logarithmic equation. $\log_4 x - 2\log_x 4 = 1$ **[5]**
$(x, y) = (1, 3)$ or $(1/4, 3/2)$	**132**	Solve the following simultaneous logarithmic equations. $\log_2(y - 1) = 1 + \log_2 x$ $2\log_3 y = 2 + \log_3 x$ **[6]**
$x = 4$ or $x = 1/4$	**133**	Solve the following logarithmic equation. $\log_2 128 - \log_2 8 = (\log_2 x)^2$ **[5]**

Answer	No.	Question
Shown: $x = (1 + \log_3 2)/(2 - \log_3 5)$	**134**	Two curves C_1 and C_2 are defined for all values of x and have respective equations $y = 9^x$ and $y = 6 \times 5^x$. Show that the x coordinate of the point of intersection is given by $(1 + \log_3 2)/(2 - \log_3 5)$. **[5]**

SECTION K: EXAM QUESTIONS (****+)

Answer	No.	Question
(a) reflection in the y axis; (b) increasing curve through (0, 3), asymptote y = 0; (c) $y = \sqrt{3}$	**135**	(a) Describe the geometric transformation which maps the graph of $y = 2^x$ onto the graph of $y = (1/2)^x$. **[1]** (b) Sketch the graph of $y = 3 \times 2^x$. **[2]** The curve with equation $y = (1/2)^x$ intersects the curve with equation $y = 3 \times 2^x$ at the point P. (c) Determine, as an exact simplified surd, the y coordinate of P. **[4]**
$x = 4$	**136**	Solve the following logarithmic equation. $2\log_2 x + \log_2(x - 1) - \log_2(5x + 4) = 1$ **[6]**
$x = \log_2 5$	**137**	The following three numbers $\log_{10} 2$, $\log_{10}(2^x - 1)$, $\log_{10}(2^x + 3)$ are consecutive terms in an arithmetic progression. Determine the value of x as an exact logarithm of base 2. **[6]**
Shown: $x = 1/(4\log_{10} 2)$	**138**	$f(x) = 2^{4x}$. Show that the solution of the equation $f(x - 1) = 5/8$ is given by $x = 1/(4\log_{10} 2)$. **[5]**
$x = 1$ or $x = 10$	**139**	Solve the following logarithmic equation, $x > 0$ (logarithms base 10). $5 \times 5^{\log x} + 5^{2-\log x} = 30$ **[6]**
Shown: $x = a^{(-1 \pm \sqrt{21})/2}$	**140**	It is given that $\Sigma_{r=1}^{3}(\log_a x)^r = \Sigma_{r=1}^{3}\log_a(x^r)$, where a and x are positive numbers such that $x \neq a$, $x \neq 1$ and $a > 1$. Show clearly that $x = a^{(-1 \pm \sqrt{21})/2}$. **[6]**
$x = 4$	**141**	Find the only real root of the following logarithmic equation. $2\log_2 x + \log_2(x - 1) - \log_2(5x + 4) = 1$ **[6]**
$x = 4\log c/\log a$, $y = -\log c/\log b$	**142**	Solve the following simultaneous equations. $a^{2x} \times b^{3y} = c^5$ $a^{3x} \times b^{2y} = c^{10}$ Give the answers in exact form in terms of log a, log b and log c. **[6]**

SECTION L: EXAM QUESTIONS (*****)

Answer	No.	Question
Shown: $(8!)^9 < (9!)^8$	**143**	Show by valid mathematical arguments that $(8!)^9 < (9!)^8$. **[5]**
2045 as worded (printed answer: 2012)	**144**	On the 1st January 2000 a rare stamp was purchased at an auction for £16384 and by the 1st January 2010 its value was four times as large as its purchase price. The future value of this stamp, £V, t years after the 1st January 2000 is modelled by $V = Ap^t$, $t \geq 0$, where A and p are positive constants. On the 1st January 1990 a different stamp was purchased for £2. The future value of this stamp, £U, t years after the 1st January 1990 is modelled by $U = Bq^t$, $t \geq 0$, where B and q are positive constants. Given further that $q = p^2$, determine the year when the two stamps will achieve the same value according to their modelling equations. **[8]**
Shown: $2(\log_a x)^n - n(n+1)\log_a x + (n-1)(n+2) = 0$	**145**	It is given that a and x are positive real numbers such that $x \neq a$, $x \neq 1$ and $a > 1$, and n is a positive integer such that $n > 1$. Show that the equation $\Sigma_{r=1}^{n}(\log_a x)^r = \Sigma_{r=1}^{n}\log_a(x^r)$ can be written as $2(\log_a x)^n - n(n+1)\log_a x + (n-1)(n+2) = 0$. **[7]**
(i) shown; (ii) (x, y) = (10000, 3) or (1000, 4)	**146**	(i) Prove that $a^{\log b} \equiv b^{\log a}$, $a > 0$, $b > 0$ (logarithms base 10). (ii) Hence, or otherwise, solve the simultaneous equations $\log x + y = 7$ and $x^y = 10^{12}$. **[8]**
k	**147**	The product operator $\prod$ is defined as $\prod_{i=1}^{k} u_i = u_1 \times u_2 \times u_3 \times \ldots \times u_k$. Given that $k \in \mathbb{N}$, use a detailed method to find the value of $\prod_{r=2}^{2^{k}-1} \log_r(r+1)$. **[5]**
k = 16	**148**	Solve the following logarithmic equation, $k > 0$, $k \neq 1$. $\log_k(8 + 4\sqrt{3}) + \log_k(8 - 4\sqrt{3}) - 1 = 0$ **[5]**
x = 1/2 or x = 8	**149**	Solve the following logarithmic equation, $x \in (0, \infty)$. $x^{\log_2 x} = 8x^2$ **[6]**
(a) shown; (b) x = 100	**150**	(a) Prove that $a^{\log b} \equiv b^{\log a}$, $a > 0$, $b > 0$. (b) Hence, or otherwise, solve the following equation (logarithms base 10). $3^{\log x} + 3 \times x^{\log 3} = 36$ **[7]**

Answer	No.	Question
Shown: $x = \log_2\varphi$	**151**	The positive solution of the quadratic equation $x^2 - x - 1 = 0$ is denoted by φ, and is commonly known as the golden section or golden number. Show, with a detailed method, that the real solution of the following exponential equation $2^x + 4^x = 8^x$ can be written in exact form as $\log_2\varphi$. **[6]**
Shown: $x^4 - x^2 - 1 = 0$	**152**	It is given that for $x > 0$, $x \neq 1$ and $y > 0$, $y \neq 1$, $\log_x y = \log_y x$ and $\log_x(x - y) = \log_y(x + y)$. Show that $x^4 - x^2 - 1 = 0$. **[7]**
Shown: $x = (8n - 7)\pi/4$, $n \in \mathbb{Z}$	**153**	$\log_{\sin x \cos x}(\sin x) \times \log_{\sin x \cos x}(\cos x) = 1/4$. Show that the solution of the above equation is given by $x = (8n - 7)\pi/4$, $n \in \mathbb{Z}$. **[8]**